

TITLE: - CPU INFORMATION

AIM: -

To write a program to find info of the processor like serial number, type, model, stepping.

ALGORITHM: -

- Initialize data segment.
- Execute CUID with EAX=0 to get Vendor ID.
- Store and display Vendor ID string.
- Execute CUID with EAX=1 to get processor signature.
- Extract Stepping, Model, and Type via bitmasking.
- Convert extracted values to ASCII hexadecimal characters.
- Display CPU Type, Model, and Stepping values.
- Terminate program execution.

PROGRAM:-

TITLE CPUINFO.ASM -- Read and Display Processor Information

```
;/*****
;
;
; Copyright (c) 1985-2025, American Megatrends International LLC.  *
;
;
; All rights reserved. Subject to AMI licensing agreement.      *
;
;
;*****/
```

```
; <AMI_FHDR_START>
```

```
;
```

```
; Header:
```

```
; Author: Muthuganesh S
```

```
; Description:
```

```
; Program to display processor vendor ID, processor type,  
; model and stepping using CUID instruction.  
;  
;<AMI_FHDR_END>  
  
.model small  
.386  
.stack 100h  
.586  
.model small ; Fixed: Changed from flat to small for 16-bit DOS  
.stack 100h  
  
;-----  
; DATA SEGMENT  
;-----  
data segment  
  
msg1 db 13,10,'Processor Vendor ID : $'  
msg2 db 13,10,'Processor Type : $'  
msg3 db 13,10,'Processor Model : $'  
msg4 db 13,10,'Processor Stepping : $'  
  
vendor db 13 dup('$')  
  
typeval db ?  
modelval db ?  
stepval db ?  
  
data ends  
;-----
```

```
;-----  
;   CODE   SEGMENT  
;-----
```

```
code segment
```

```
assume cs:code, ds:data
```

```
start:
```

```
main proc
```

```
    mov ax, data
```

```
    mov ds, ax
```

```
    call customfunction
```

```
    mov ax, 4C00h
```

```
    int 21h
```

```
main endp
```

```
customfunction proc
```

```
    ; --- Get Vendor ID ---
```

```
    mov eax, 0
```

```
    cpuid
```

```
mov dword ptr vendor[0], ebx
mov dword ptr vendor[4], edx
mov dword ptr vendor[8], ecx
```

```
mov ah,09h
lea dx,msg1
int 21h
```

```
mov ah,09h
lea dx,vendor
int 21h
```

```
; --- Get CPU Info ---
mov eax,1
cpuid
```

```
; Get Stepping (Bits 0-3)
mov ebx,eax
and bl,0Fh
mov stepval,bl
```

```
; Get Model (Bits 4-7)
mov ebx,eax
shr ebx,4    ; Fixed: Shift 32-bit register ebx
and bl,0Fh
mov modelval,bl
```

```
; Get Type (Bits 12-13)
mov ebx,eax
shr ebx,12   ; Fixed: Shift 32-bit register ebx
```

```
and bl,03h  
mov typeval,bl
```

```
; --- Print Results ---
```

```
; Type  
mov ah,09h  
lea dx,msg2  
int 21h  
mov dl,typeval  
add dl,30h  
mov ah,02h  
int 21h
```

```
; Model  
mov ah,09h  
lea dx,msg3  
int 21h  
mov dl,modelval  
add dl,30h    ; Prints symbol if > 9  
mov ah,02h  
int 21h
```

```
; Stepping  
mov ah,09h  
lea dx,msg4  
int 21h  
mov dl,stepval  
add dl,30h    ; Prints symbol if > 9  
mov ah,02h
```

int 21h

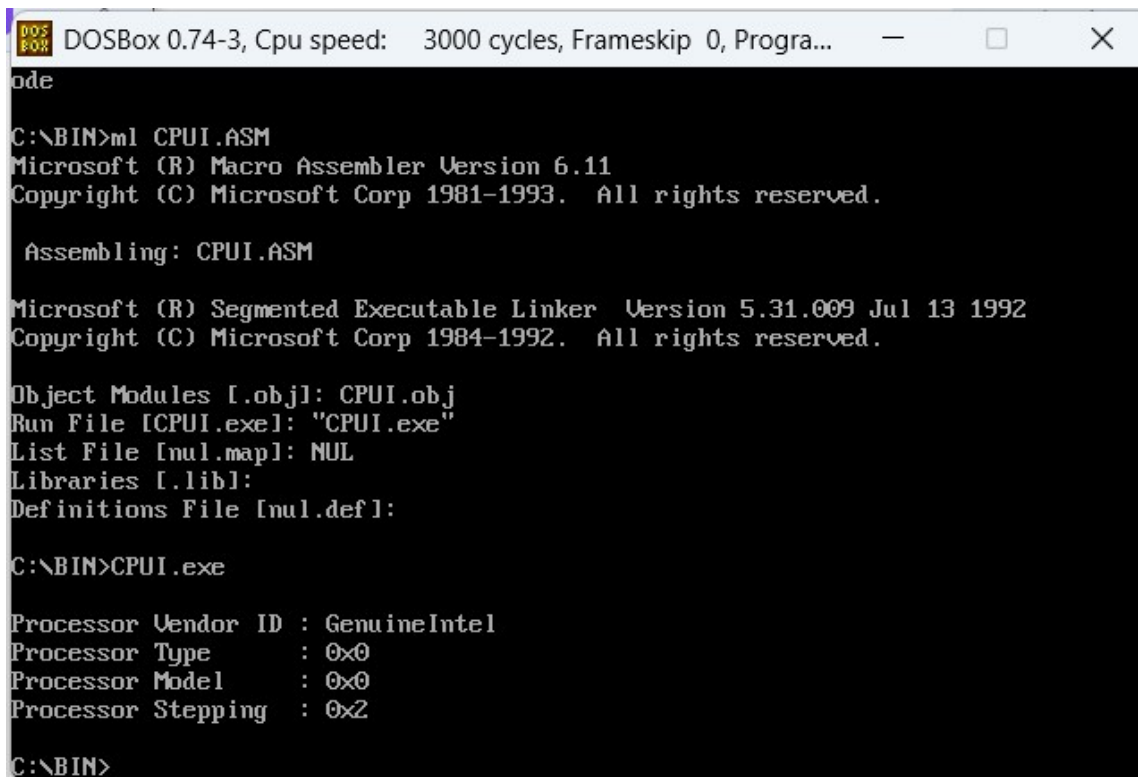
ret

customfunction endp

code ends

end start

OUTPUT: -



```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Progra...
ode
C:\BIN>ml CPUI.ASM
Microsoft (R) Macro Assembler Version 6.11
Copyright (C) Microsoft Corp 1981-1993. All rights reserved.

Assembling: CPUI.ASM

Microsoft (R) Segmented Executable Linker Version 5.31.009 Jul 13 1992
Copyright (C) Microsoft Corp 1984-1992. All rights reserved.

Object Modules [.obj]: CPUI.obj
Run File [CPUI.exe]: "CPUI.exe"
List File [nul.map]: NUL
Libraries [.lib]:
Definitions File [nul.def]:

C:\BIN>CPUI.exe

Processor Vendor ID : GenuineIntel
Processor Type      : 0x0
Processor Model     : 0x0
Processor Stepping  : 0x2

C:\BIN>
```